Bauhaus-Universität Weimar

JOHNS HOPKINS UNIVERSITY

From FEM to Neural Operators: Scientific Machine Learning for Computational Mechanics

Parent Element:
Basis Functions, Jacobians and Gauss Quadrature

Dr-Ing Jorge Alberto López Zermeño

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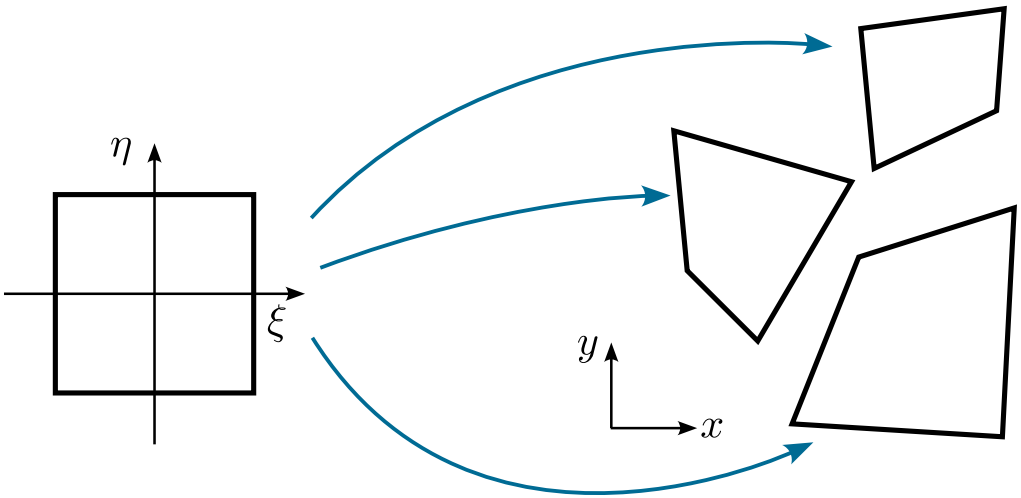
Why a parent element?

- Standardize on a **single reference domain**; map to real geometry.
- Compute shapes, derivatives, and integrals in the parent space.
- Clean, reusable element routines across arbitrary meshes.



1D Lagrange bases ($\xi \in [-1, 1]$)

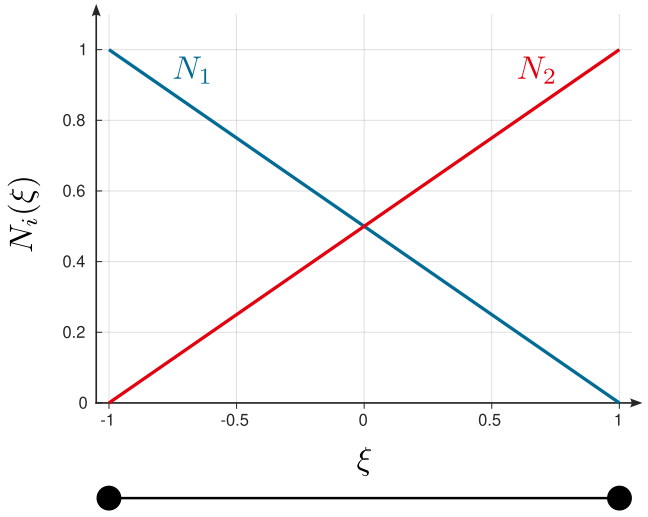
General formula

$$N_a(\xi) = \prod_{b \neq a} \frac{\xi - \xi_b}{\xi_a - \xi_b}, \quad N_a(\xi_b) = \delta_{ab}$$

Linear Basis Functions ($p = 1$)

$N_1 = \frac{1-\xi}{2}$

$N_2 = \frac{1+\xi}{2}$

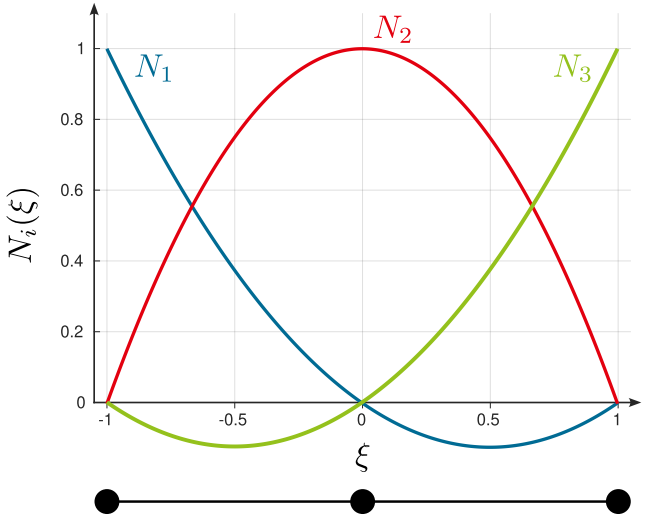


Quadratic Basis Functions ($p = 2$)

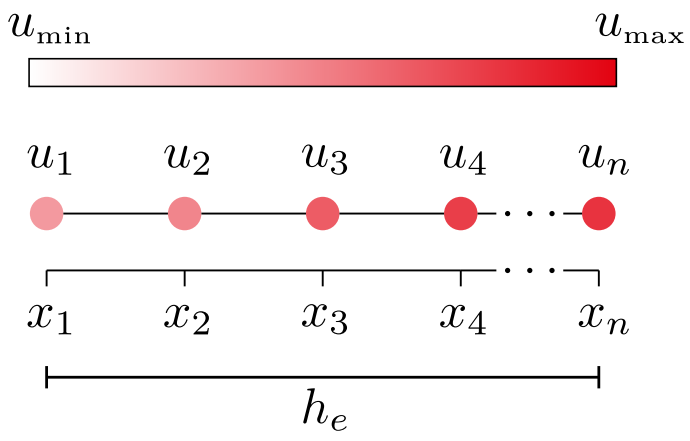
$N_1 = \frac{1}{2}\xi(\xi - 1)$

$N_2 = 1 - \xi^2$

$N_3 = \frac{1}{2}\xi(\xi + 1)$



1D isoparametric mapping



- Same N_i for interpolating $x(\xi)$ and $u(\xi)$

$$x(\xi) = \sum_i N_i x_i, \quad u(\xi) = \sum_i N_i u_i,$$

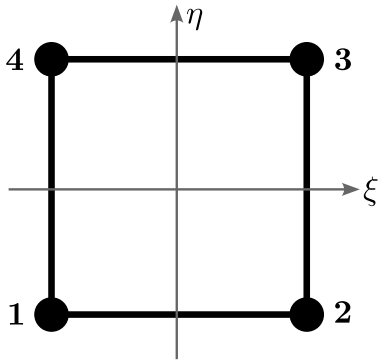
- Jacobian: Represents how $d\xi$ scales into dx

$$J = \frac{dx}{d\xi} = \sum_i x_i N_i' \quad \frac{du}{dx} = \frac{1}{J} \sum_i u_i N_i'$$

The Jacobian $J(\xi)$ corrects the integral over ξ, so we still integrate over the physical length x.

- Integral: $\int g(x)\, dx = \int g(x(\xi))\, J(\xi)\, d\xi$
- Check: $J(\xi) > 0$

Q4 Element

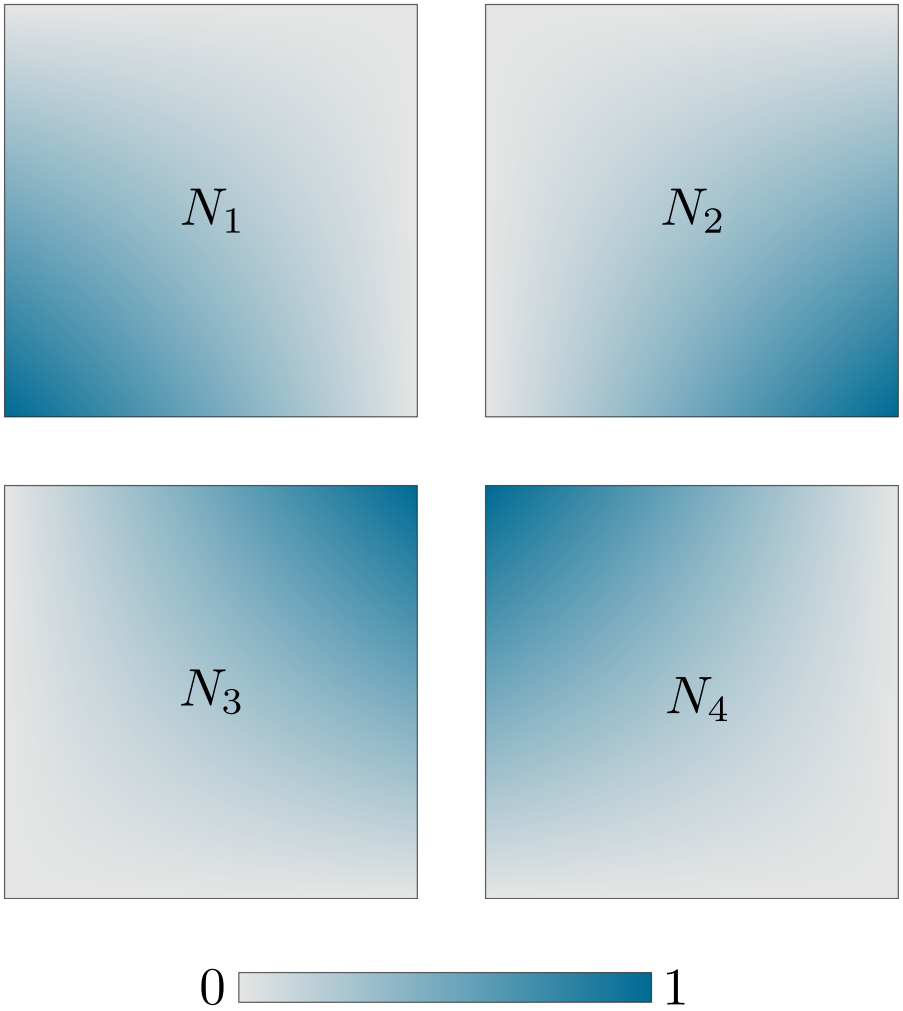


$$N_1 = \frac{1}{4}(1-\xi)(1-\eta),$$
$$N_2 = \frac{1}{4}(1+\xi)(1-\eta),$$
$$N_3 = \frac{1}{4}(1+\xi)(1+\eta),$$
$$N_4 = \frac{1}{4}(1-\xi)(1+\eta)$$

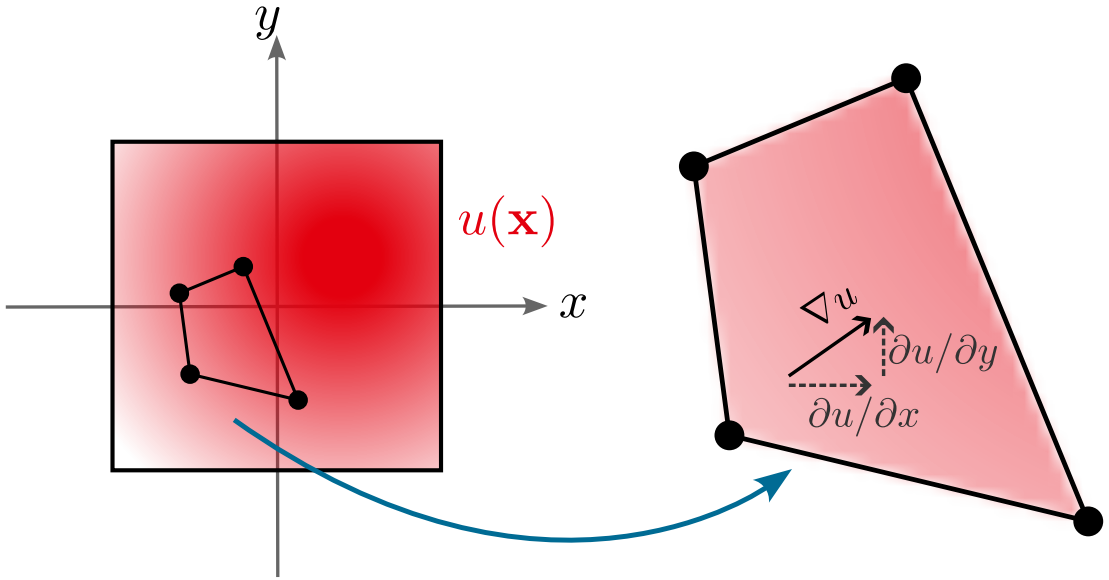
$$x(\xi,\eta) = \sum N_i x_i,\ y(\xi,\eta) = \sum N_i y_i$$

$$\mathbf{J} = \begin{bmatrix} \sum x_i\,\partial_\xi N_i & \sum y_i\,\partial_\xi N_i \\ \sum x_i\,\partial_\eta N_i & \sum y_i\,\partial_\eta N_i \end{bmatrix},$$

$$\det \mathbf{J} > 0$$



Physical gradients via chain rule



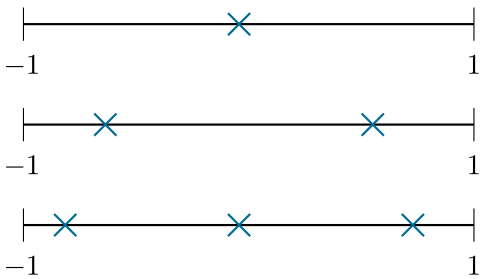
$$\nabla_\xi N_i = \begin{bmatrix} \partial_\xi N_i \\ \partial_\eta N_i \end{bmatrix}, \quad \nabla N_i = \mathbf{J}^{-1} \nabla_\xi N_i, \quad \nabla u = \sum_i \nabla N_i \, u_i$$

- $\mathbf{J}$ maps small parent areas to physical areas:
  $d\Omega = |\mathbf{J}| d\xi d\eta \Rightarrow \iint g \ \det \mathbf{J} \, d\xi \, d\eta$
- $\mathbf{J}^{-1}$ maps parent gradients to physical gradients: $\nabla = \mathbf{J}^{-1} \nabla_\xi$
- Avoid near-zero $\det \mathbf{J}$ (distortion)

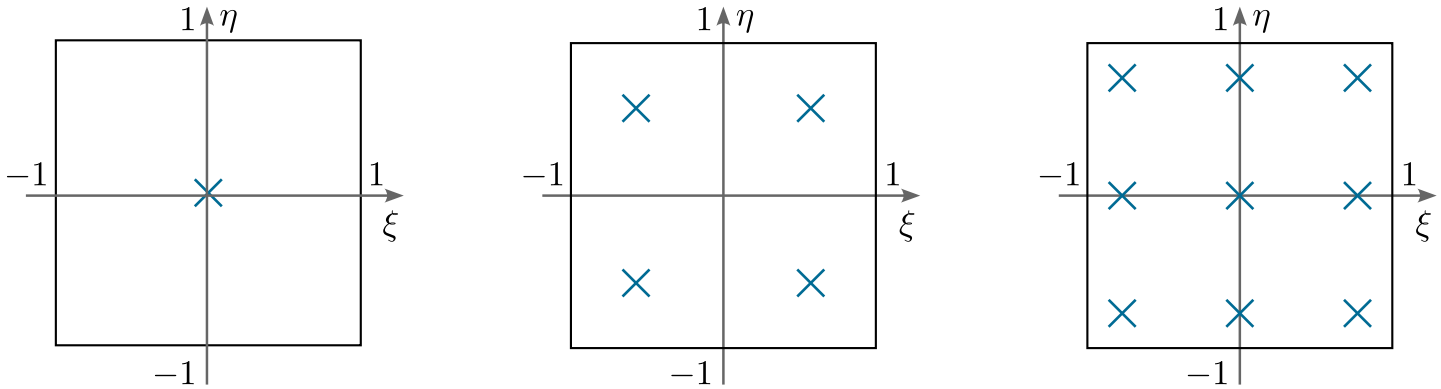
Gauss–Legendre on $[-1, 1]$

$$\int_{-1}^{1} f(\xi)\, d\xi \approx \sum_k w_k f(\xi_k), \qquad \int g(x)\, dx \approx \sum_k w_k\, g(x(\xi_k))\, J(\xi_k)$$

- 1-pt: $\xi = \{0\}$, $w = \{2\}$
- 2-pt: $\xi = \{\pm 1/\sqrt{3}\}$, $w = \{1, 1\}$
- 3-pt: $\xi = \{0, \pm\sqrt{3/5}\}$, $w = \{8/9, 5/9, 5/9\}$



$$\iint f(\xi, \eta)\, d\xi\, d\eta \approx \sum_{k,\ell} w_k w_\ell\, f(\xi_k, \eta_\ell), \qquad \int_{\Omega_e} g\, d\Omega \approx \sum_{k,\ell} w_k w_\ell\, g(\xi_k, \eta_\ell)\ \det \mathbf{J}(\xi_k, \eta_\ell)$$



Summary and outlook

Engineering checks

- Check $\det \mathbf{J}(\xi, \eta) > 0$ on every element (no inverted elements).
- Quadrature: choose Gauss order so that the rule is exact for the polynomial degree of the integrand.
- Mesh quality: avoid elements with very small $|\det \mathbf{J}|$ or very large $\|\mathbf{J}^{-1}\|$.
- Reuse the same parent-element, $\mathbf{J}$, and quadrature routines for all elements.

Relevance for PINNs and Neural Operators

- The same parent-element mapping and Jacobian appear when evaluating PDE residuals at quadrature points.
- In weak-form / variational PINNs, domain and boundary integrals in the loss use exactly these $\mathbf{J}$ and Gauss rules.
- Othe scientific computing approaches are often trained on FEM data computed with these interpolation and quadrature choices.